The documentation and process conversion measures necessary to comply with this document shall be completed by 22 August 2005.

INCH-POUND

MIL-PRF-19500/454F
20 May 2005
SUPERSEDING
MIL-PRF-19500/454E
7 October 2002

* PERFORMANCE SPECIFICATION SHEET

* SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER
TYPES: 2N5660, 2N5660U3, 2N5661, 2N5661U3, 2N5662, AND, 2N5663, JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, power transistors. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. For types 2N5660 and 2N5661, see figure 1 (similar to TO-66). For types 2N5662 and 2N5663, see figure 2 (similar to TO-5). For types 2N5660U3, and 2N5661U3, see figure 3 (U3).

1.3 Maximum ratings. T_C = +25°C, unless otherwise specified.

Type	P_T T_A = +25°C (1) (2)	P_T T_C = +100°C (1) (2)	$R_{\theta JA}$	$R_{\theta JC}$	V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	V_{CER}	T_J and T_{STG}
	W	W	°C/W	°C/W	V dc	V dc	V dc	A dc	A dc	V dc	°C
2N5660	2.0	20	87.5	5	250	200	6.0	0.5	2.0	250	-65 to +200
2N5661	2.0	20	87.5	5	400	300	6.0	0.5	2.0	400	-65 to +200
2N5662	1.0	15	175	6.7	250	200	6.0	0.5	2.0	250	-65 to +200
2N5663	1.0	15	175	6.7	400	300	6.0	0.5	2.0	400	-65 to +200
2N5660U3		20		4.5	250	200	6.0	0.5	2.0	250	-65 to +200
2N5661U3		20		4	400	300	6.0	0.5	2.0	400	-65 to +200

(1) For derating, see figures 4 through 8.
(2) For thermal curves, see figures 9 through 16.

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to semiconductor@.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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* 1.4 Primary electrical characteristics.

Limits	h_{FE2} (1) I_C = 0.5 A dc V_{CE} = 5 V dc		$\lvert h_{fe} \rvert$ I_C = 0.1 A dc V_{CE} = 5 V dc f = 10 MHz	$V_{BE(sat)}$ (1) I_C = 1.0 A dc I_B = 0.1 A dc	$V_{CE(sat)}$ (1) I_C = 1 A dc I_B = 0.1 A dc	Pulse response		
						t_{on} I_C = 0.5 A dc	t_{off} I_C = 0.5 A dc	
	2N5660 2N5660U3 2N5662	2N5661, 2N5661U3 2N5663					2N5660 2N5660U3 2N5662	2N5661, 2N5661U3 2N5663
				V dc	V dc	µs	µs	µs
Min	40	25	2					
Max	120	75	7	1.2	0.4	0.25	0.85	1.2

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3. Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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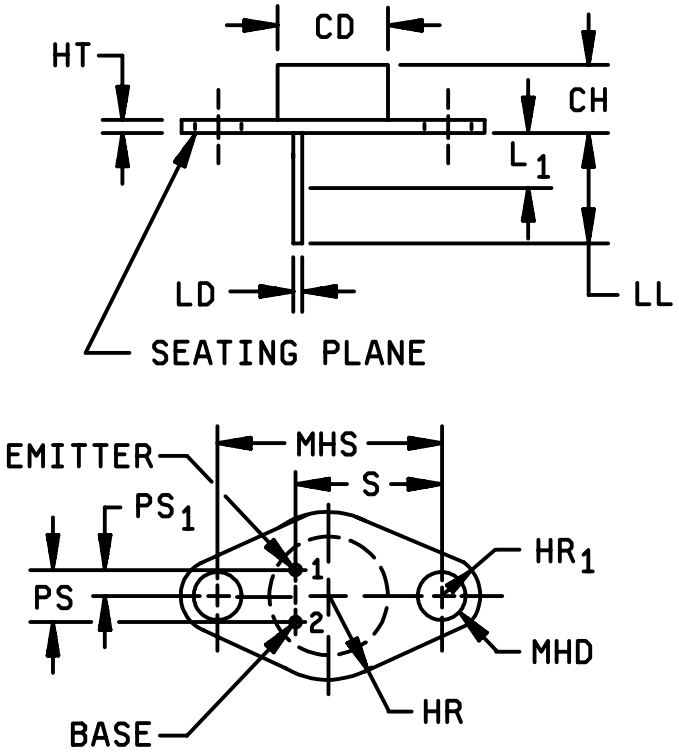


FIGURE 1. Physical dimensions, 2N5660 and 2N5661, (similar to TO-66).

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.470	.500	11.94	12.70	7
CH	.250	.340	6.35	8.64	
HR		.350		8.89	
HR_1	.115	.145	2.92	3.68	4
HT	.050	.075	1.27	1.91	
LD	.028	.034	0.71	0.86	4, 6
LL	.360	.500	9.14	12.70	4
L_1		.050		1.27	4,6
MHD	.142	.152	3.61	3.86	4
MHS	.958	.962	24.33	24.43	
PS	.190	.210	4.83	5.33	3
PS_1	.093	.107	2.36	2.72	3
s	.570	.590	14.48	14.99	3

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. These dimensions should be measured at points .050 inch (1.27 mm) +.005 inch (0.13 mm) -.000 inch (0.00 mm) below seating plane. When gauge is not used, measurement will be made at the seating plane.
4. Two places.
5. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.
6. Lead diameter shall not exceed twice LD within L_1.
7. Body contour is optional within zone defined by CD.
8. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.
9. Lead 1 is emitter, lead 2 is base, and case is collector.

FIGURE 1. Physical dimensions, 2N5660 and 2N5661, (similar to TO-66) – Continued.

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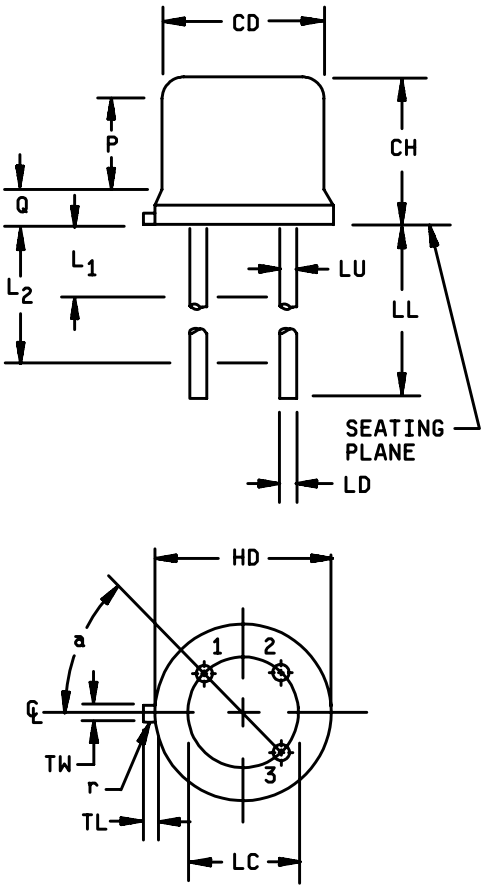


FIGURE 2. Physical dimensions, 2N5662 and 2N5663, (similar to TO-5).

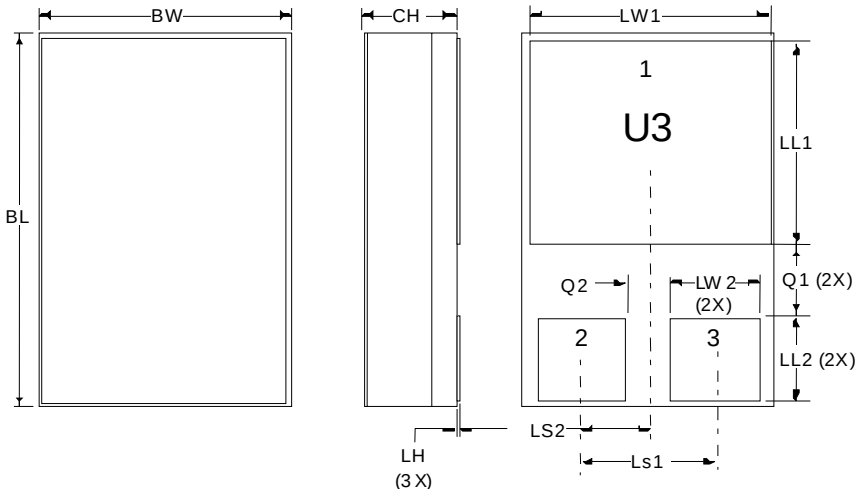
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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.355	7.75	9.02	
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	
LC	.200 TP		5.08 TP		6
LD	.016	.021	0.41	0.53	7
LL	1.500	1.750	38.10	44.45	7
LU	.016	.019	0.407	0.482	7
L1		.050		1.27	7
L2	.250		6.35		7
TL	.029	.045	0.74	1.14	3
TW	.028	.034	0.712	0.863	9
P	.100		2.54		
Q		.050		1.27	4
r		.010		0.25	10
α	45°TP		45°TP		6

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Symbol TL is measured from HD maximum.
4. Details of outline in this zone are optional.
5. Symbol CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gauge plane .054 inch (1.37 mm) +.001 inch (0.03 mm) - .000 inch (0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of TP relative to tab. Device may be measured by direct methods or by gauge.
7. Symbol LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum.
8. Lead number three is electrically connected to case.
9. Beyond r maximum, TW shall be held for a minimum length of .011 inch (0.28 mm).
10. Symbol r applied to both inside corners of tab.
11. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.
12. Lead 1 is emitter, lead 2 is base, and lead 3 is collector.

FIGURE 2. Physical dimensions, 2N5662 and 2N5663, (similar to TO-5) – Continued.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.395	.405	10.04	10.28
BW	.291	.301	7.40	7.64
CH	.1085	.1205	2.76	3.06
LH	.010	.020	0.25	0.51
LW1	.281	.291	7.14	7.39
LW2	.090	.100	2.29	2.54
LL1	.220	.230	5.59	5.84
LL2	.115	.125	2.93	3.17
LS1	.150 BSC		3.81 BSC	
LS2	.075 BSC		1.91 BSC	
Q1	.030		0.762	
Q2	.030		0.762	
TERM 1	Collector			
TERM 2	Base			
TERM 3	Emitter			

* FIGURE 3. Physical dimensions, 2N5660U3 and 2N5661U3 (U3)

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3. REQUIREMENTS

* 3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and figure 1 (similar to TO-66), figure 2 (similar to TO-5), and figure 3 (U3) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of finish is desired, it shall be specified in the contract or order (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

* 4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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4.3 Screening (JANTX and JANTXV levels). Screening shall be in accordance with table IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement
	JANTX JANTXV levels
3c	Thermal impedance, method 3131 of MIL-STD-750
7	Optional
9	I_{CES1}
10	48 hours minimum
11	I_{CES1} and h_{FE2}; ΔI_{CES1} = 100 percent of initial value or 20 nA, dc whichever is greater.
12	See 4.3.1 80 hours minimum
13	Subgroup 2 of table I herein; ΔI_{CES1} = 100 percent of initial value or 20 nA dc; Δh_{FE2} = ±25 percent of initial value.
14	Required

* 4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: V_{CB} = 10 - 30 V dc, T_A = room ambient as defined in 4.5 of MIL-STD-750. Power shall be applied to the device to achieve the required junction temperature, T_J = 135°C minimum and a minimum power dissipation = 75 percent of max P_T as defined in 1.3.

* 4.3.2 Thermal impedance measurements. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD} (and V_C where appropriate). The thermal impedance limit used in screen 3c of 4.3 shall comply with the thermal impedance graph on figures 9 through 16 (less than or equal to the curve value at the same t_H time) and shall be less than the process determined statistical maximum limit as outlined in method 3131.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, subgroup 2 herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified below. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) and delta requirements JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with table I, subgroup 2, and table III herein.

* 4.4.2.1 Group B inspection, table VIb (JAN, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action.

Step	Method	Condition
1	1039	Steady-state life: Test condition B, 1,000 hours minimum, V_{CB} = 10 V dc, power shall be applied to achieve T_J = +150°C minimum using a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3. n = 45 devices, c = 0.
2	1039	HTRB: Test condition A, 48 hours minimum. n = 45 devices, c = 0.
3	1032	High-temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0.

4.4.2.2 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection, Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, 4.4.3.1 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements shall be in accordance with table I, and subgroup 2 herein.

4.4.3.1 Group C inspection, table VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition A (tension) for 2N5660 and 2N5661; weight = 3 pounds; t = 15s, not applicable for U3 devices.
C2	2036	Test condition E (lead fatigue) for 2N5662 and 2N5663.
C5	3131	See 4.4.5, n = 22, c = 0.
C6		Not applicable.

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4.4.3.2 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IX of MIL-PRF-19500, and table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2, herein.

4.4.5 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power application shall be 0.5 A dc for 2N5660, 2N5661, 2N5660U3, and 2N5661U3, and 0.4 A dc for 2N5662 and 2N5663.

b. Collector to emitter voltage magnitude shall be 20 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be $+25°C \leq T_R \leq +75°C$ and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to header.

f. Maximum limit of $R_{\theta JC}$ shall be 5°C/W for 2N5660 and 2N5661, 4.5°C/W 2N5660U3, 4°C/W 2N5661U3, and 6.67°C/W for 2N5662 and 2N5663.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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* TABLE I. Group A inspection

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical examination	2071	n = 45 devices, c = 0				
Solderability 3/	2026	n = 15 leads, c = 0				
Resistance to solvent 3/ 4/	1022	n = 15 devices, c = 0				
Temp cycling 3/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Hermetic seal	1071	n = 22 devices, c = 0				
Fine leak						
Gross leak						
Electrical measurements		See table I, subgroup 2				
Bond strength 3/	2037	Precondition T_A = +250°C at t = 24 hrs or T_A = +300°C at t = 2 hrs, n = 11 wires, c = 0				
Subgroup 2						
Thermal impedance	3131	See 4.3.2	$Z_{\theta JX}$			°C/W
Breakdown voltage, collector to emitter	3011	Pulsed (see 4.5.1) I_C = 10 mA dc, bias condition D	$V_{(BR)CEO}$			V dc
2N5660, U3, 2N5662				200		
2N5661, U3, 2N5663				300		
Breakdown voltage, collector to base	3001	Bias condition D, I_C = 10 mA dc, pulsed (see 4.5.1)	$V_{(BR)CBO}$			V dc
2N5660, U3, 2N5662				250		
2N5661, U3, 2N5663				400		
Breakdown voltage, emitter to base	3026	Bias condition D, I_E = 10 µA dc	$V_{(BR)EBO}$	6		V dc
Collector to emitter cutoff current	3041	Bias condition C	I_{CES1}			
2N5660, U3, 2N5662		V_{CE} = 200 V dc			0.2	µA dc
2N5661, U3, 2N5663		V_{CE} = 300 V dc			0.2	µA dc
Collector to base cutoff current	3036	Bias condition D	I_{CBO}			
2N5660, U3, 2N5662		V_{CB} = 200 V dc			0.1	µA dc
		V_{CB} = 250 V dc			1.0	mA dc
2N5661, U3, 2N5663		V_{CB} = 300 V dc			0.1	µA dc
		V_{CB} = 400 V dc			1.0	mA dc

See footnotes at end of table.

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* TABLE I. Group A inspection. - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 – Continued						
Base emitter voltage (saturated)	3066	Test condition A, pulsed (see 4.5.1), I_C = 1 A dc, I_B = 0.1 A dc	$V_{BE(sat1)}$		1.2	V dc
Base emitter voltage (saturated)	3066	Test condition A, pulsed (see 4.5.1), I_C = 2 A dc, I_B = 0.4 A dc	$V_{BE(sat2)}$		1.5	V dc
Collector to emitter voltage (saturated)	3071	Pulsed (see 4.5.1), I_C = 1 A dc, I_B = 0.1 A dc	$V_{CE(sat1)}$		0.4	V dc
Collector to emitter voltage (saturated)	3071	Pulsed (see 4.5.1), I_C = 2 A dc, I_B = 0.4 A dc	$V_{CE(sat2)}$		0.8	V dc
Forward current transfer ratio	3076	V_{CE} = 2.0 V dc, I_C = 50 mA dc, pulsed (see 4.5.1)	h_{FE1}			
2N5660, U3, 2N5662				40		
2N5661, U3, 2N5663				25		
Forward current transfer ratio	3076	V_{CE} = 5.0 V dc, I_C = 0.5 A dc, pulsed (see 4.5.1)	h_{FE2}			
2N5660, U3, 2N5662				40	120	
2N5661, U3, 2N5663				25	75	
Forward current transfer ratio	3076	V_{CE} = 5.0 V dc, I_C = 1 A dc, pulsed (see 4.5.1)	h_{FE3}	15		
Forward current transfer ratio	3076	V_{CE} = 5.0 V dc, I_C = 2 A dc, pulsed (see 4.5.1)	h_{FE4}	5		
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition C	I_{CES2}			
2N5660, U3, 2N5662		V_{CE} = 200 V dc			9	µA dc
2N5661, U3, 2N5663		V_{CE} = 300 V dc			9	µA dc
Low temperature operation:		T_A = -55°C				
Forward current transfer ratio	3076	V_{CE} = 5.0 V dc, I_C = 0.5 A dc, pulsed (see 4.5.1)	h_{FE5}			
2N5660, U3, 2N5662				15		
2N5661, U3, 2N5663				10		

See footnotes at end of table.

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* TABLE I. Group A inspection. - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 5 V dc, I_C = 0.1 A dc, f = 10 MHz	$\|h_{fe}\|$	2	7	
Pulse response						
Turn-on time	3251	Test condition A, V_{CC} = 100 V dc, I_C = 0.5 A dc	t_{on}			
2N5660, U3, 2N5662		See figure 17			0.25	µs
2N5661, U3, 2N5663		See figure 18			0.25	µs
Turn-off time	3251	Test condition A, V_{CC} = 100 V dc, I_C = 0.5 A dc	t_{off}			
2N5660, U3, 2N5662		See figure 17			0.85	µs
2N5661, U3, 2N5663		See figure 18			1.2	µs
Open circuit output capacitance	3236	V_{CB} = 10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{obo}		45	pF
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = +100°C, power application time ≥ 1 second, 1 cycle, $t_r + t_f$ = 10 µs, (2N5660 2N5660U3, 2N5661, and 2N5661U3, see figure 19, 2N5662 and 2N5663, see figure 20)				
Test 1						
2N5660, U3, 2N5661, U3		V_{CE} = 10 V dc, I_C = 2 A dc				
2N5662, 2N5663		V_{CE} = 7.5 V dc, I_C = 2 A dc				
Test 2						
2N5660, U3, 2N5661, U3		V_{CE} = 40 V dc, I_C = 500 mA dc				
2N5662, 2N5663		V_{CE} = 25 V dc, I_C = 600 mA dc				
Test 3						
2N5660, U3		V_{CE} = 200 V dc, I_C = 36 mA dc				
2N5662		V_{CE} = 200 V dc, I_C = 27 mA dc				
Test 4						
2N5661, U3		V_{CE} = 300 V dc, I_C = 19 mA dc				
2N5663		V_{CE} = 300 V dc, I_C = 14 mA dc				

See footnotes at end of table.

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* TABLE I. Group A inspection. - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 5 – Continued						
Safe operating area (clamped inductive)		T_C = +100°C, $t_r + t_f \leq 10$ µs duty cycle ≤ 2 percent, I_C = 2 A dc, $t_p \approx 4$ ms, V_{CE} = 25 V dc (see figures 21 and 22)				
2N5660, U3, 2N5662 2N5661, U3, 2N5663		Clamp voltage = 200, +0, -5 V dc Clamp voltage = 300, +0, -5 V dc				
Electrical measurements		See table I, subgroup 2 herein.				

1/ For sampling plan, see MIL-PRF-19500.
2/ For resubmission of failed subgroup 1, double the sample size of the failed test or sequence of tests.
3/ Separate samples may be used.
4/ Not required for laser marked devices.

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* TABLE II. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Qualification conformance inspection
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Thermal shock (temperature cycling)	1051	500 cycles, condition D	
Electrical measurements		See table I, subgroup 2	
Subgroup 2			45 devices, c = 0
Steady-state dc blocking life	1039 or 1049	Condition A, 500 hours	
Electrical measurements		See table I, subgroup 2, I_{CBO} test only at V_{CBO} maximum rating (see 1.3).	
Subgroups 4			Sample size N/A
Thermal impedance, thermal resistance curves		Each supplier shall submit their qual-lot average and design thermal impedance curves to the qualifying activity. In addition, the optimal test conditions and thermal impedance limit shall be provided to the qualifying activity in the qualification report.	
Subgroup 5			15 devices, c = 0
Barometric pressure (reduced)	1001	Normal mounting pressure = 8 mm Hg ±2 mm Hg for 60 s (minimum)	
Electrical measurements		See table I, subgroup 2	
Subgroups 6			3 devices c = 0
ESD	1020		
Subgroup 8			45 devices, c = 0
Reverse stability	1033	Condition A for devices ≥ 400 V, condition B for devices < 400 V.	

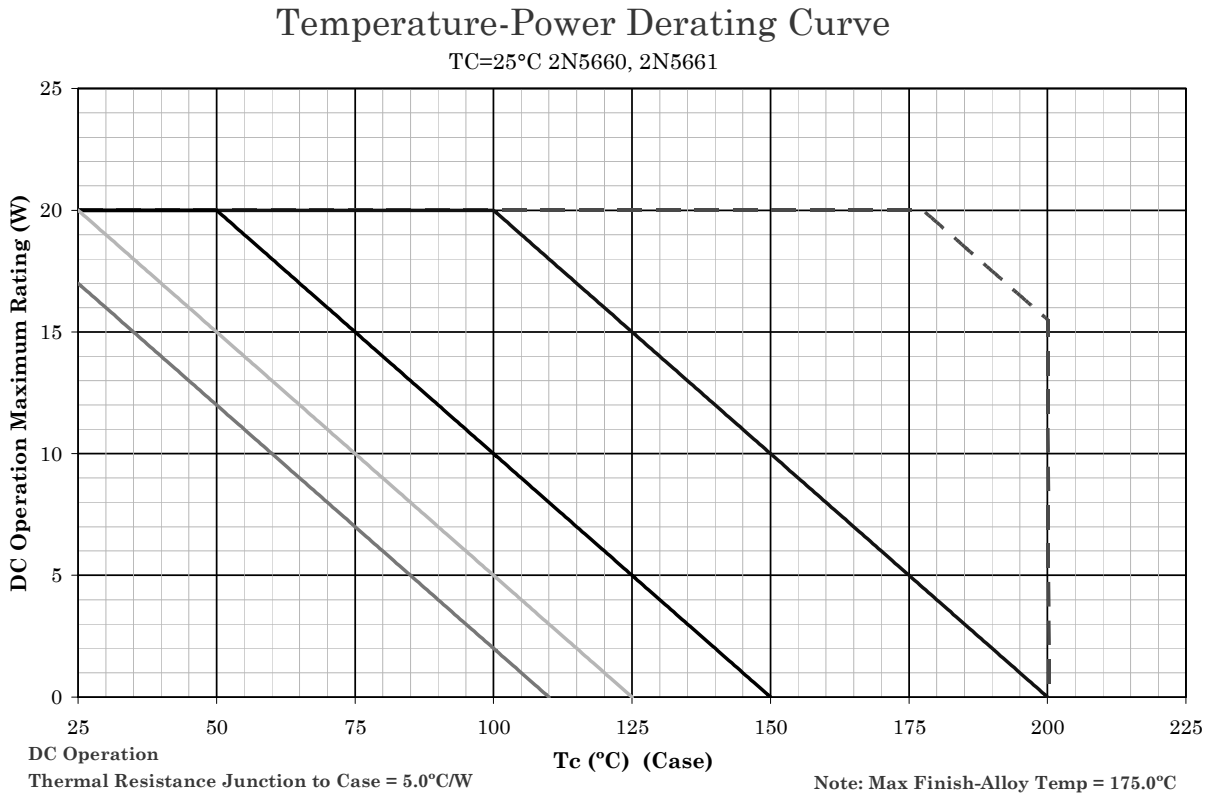
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* TABLE III. Delta measurements. 1/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector to emitter cutoff current 2N5660, 2N5662 2N5661, 2N5663	3041	Bias condition C V_{CE} = 200 V dc V_{CE} = 300 V dc	ΔI_{CES1}	100 percent of initial value or 20 nA dc, whichever is greater.		
2.	Forward current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 0.5 A dc, pulsed (see 4.5.1)	Δh_{FE2}	±25 percent change in initial recorded value.		
3.	Thermal impedance	3131	See 4.4.5	$\Delta Z_{\theta JC}$	±10 percent change in initial recorded value.		°C/W

1/ The delta measurements for 4.4.2 are as follows:
a. Step 1, see table III, steps 1, 2, and 3.
b. Step 2, see table III, steps 1, 2, and 3.
c. Step 3, see table III, steps 1 and 2.

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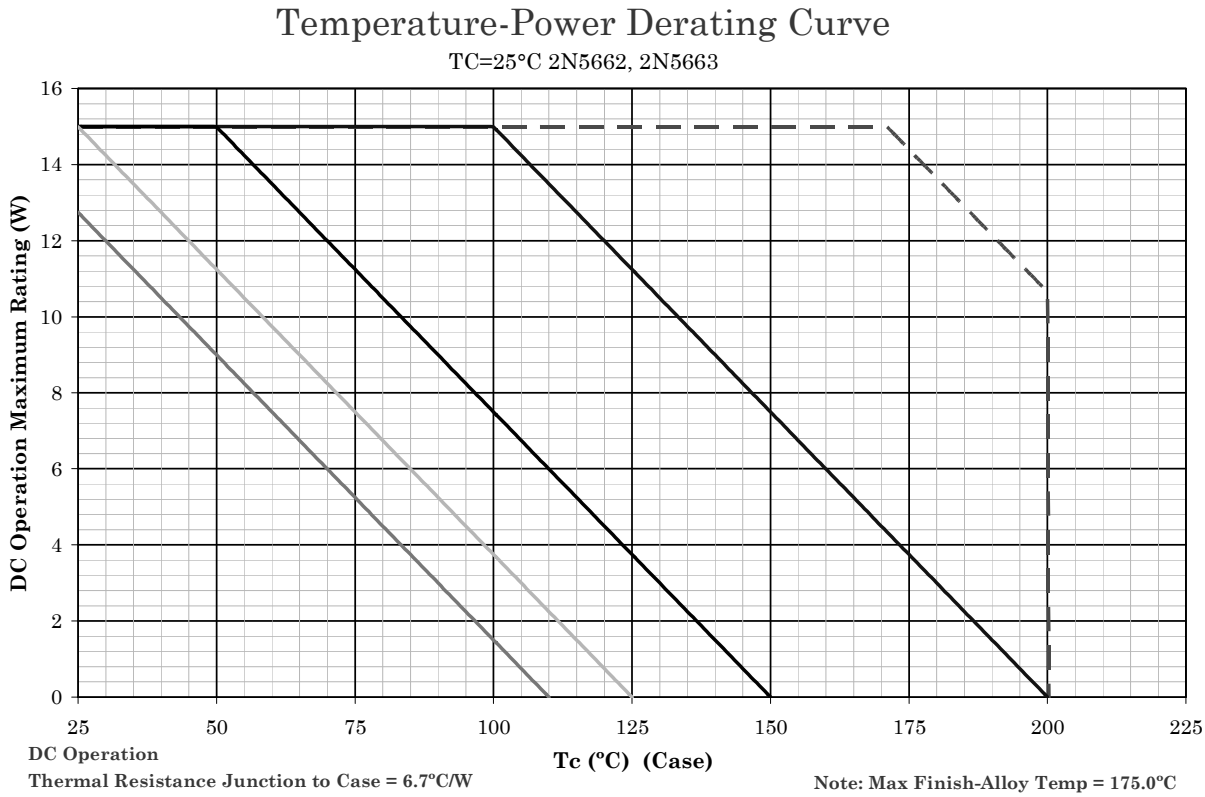


NOTES:

1. Maximum theoretical derate design curve. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 4. Derating for 2N5660 and 2N5661 (similar to TO-66).

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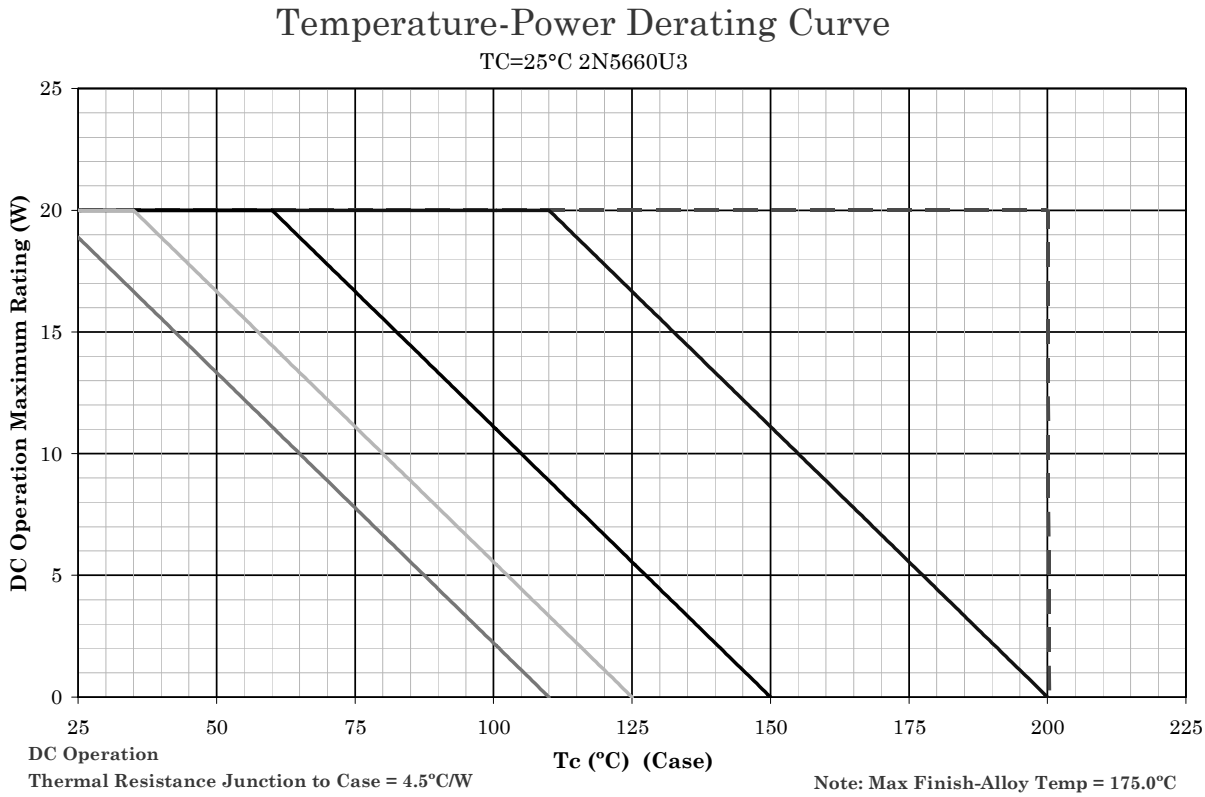


NOTES:

1. Maximum theoretical derate design curve. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 5. Derating for 2N5662 and 2N5663 (similar to TO-5).

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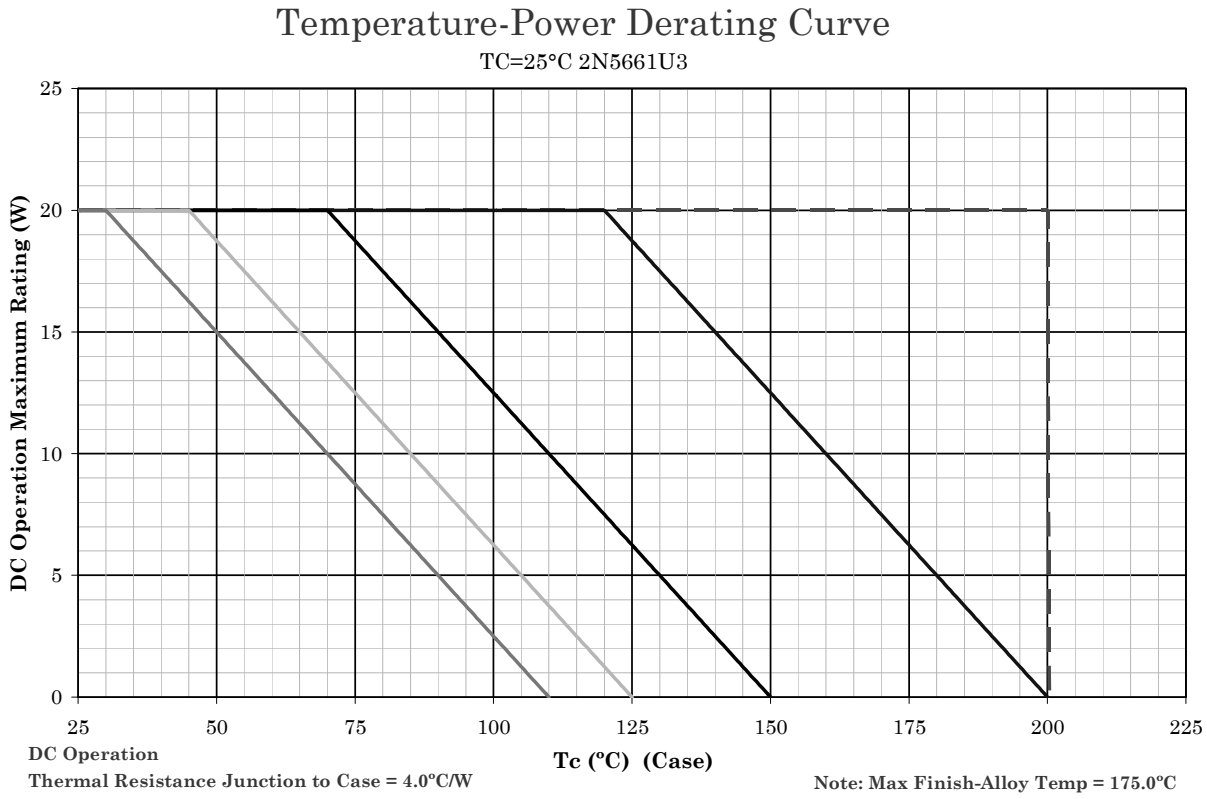


NOTES:

1. Maximum theoretical derate design curve. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 6. Derating for 2N5660U3 (U3).

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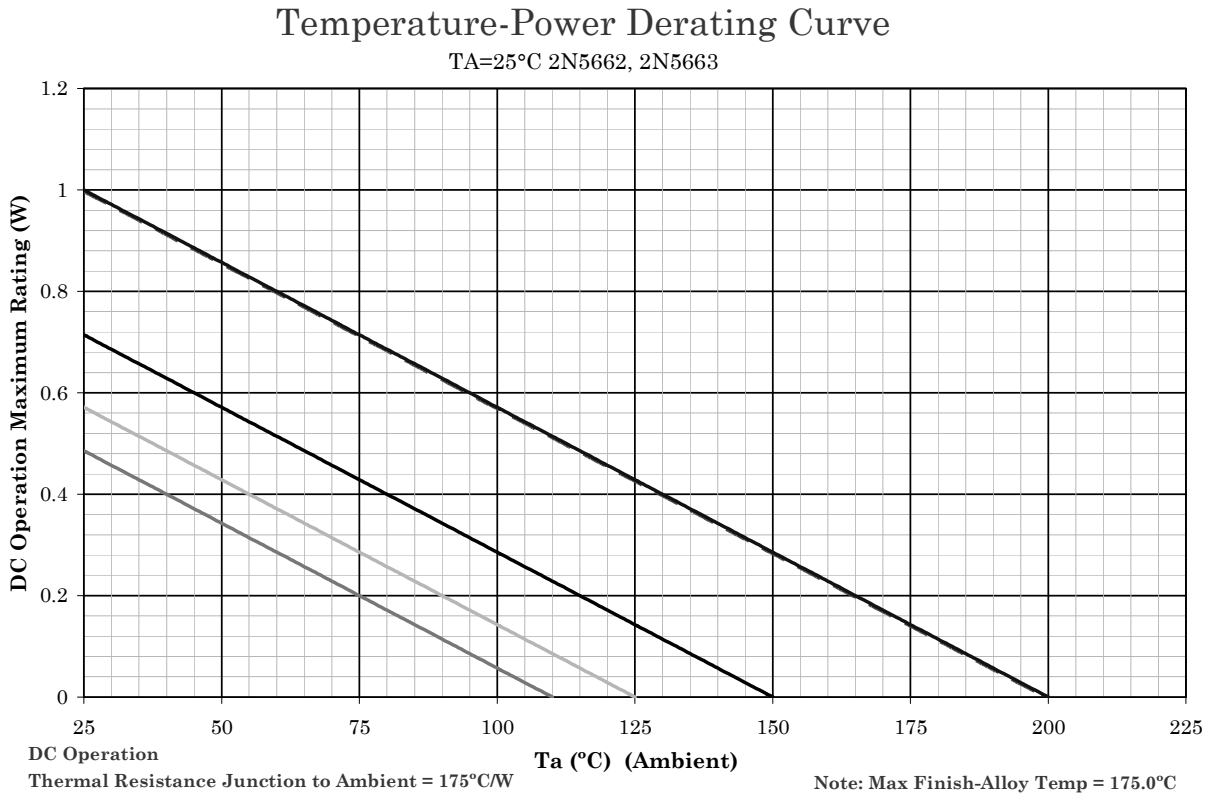


NOTES:

1. Maximum theoretical derate design curve. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 7. Derating for 2N5661U3 (U3).

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NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See paragraph 1.3 herein)
2. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
3. Derate design curve chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 8. Derating for 2N5662, 2N5663 (similar to TO-5).

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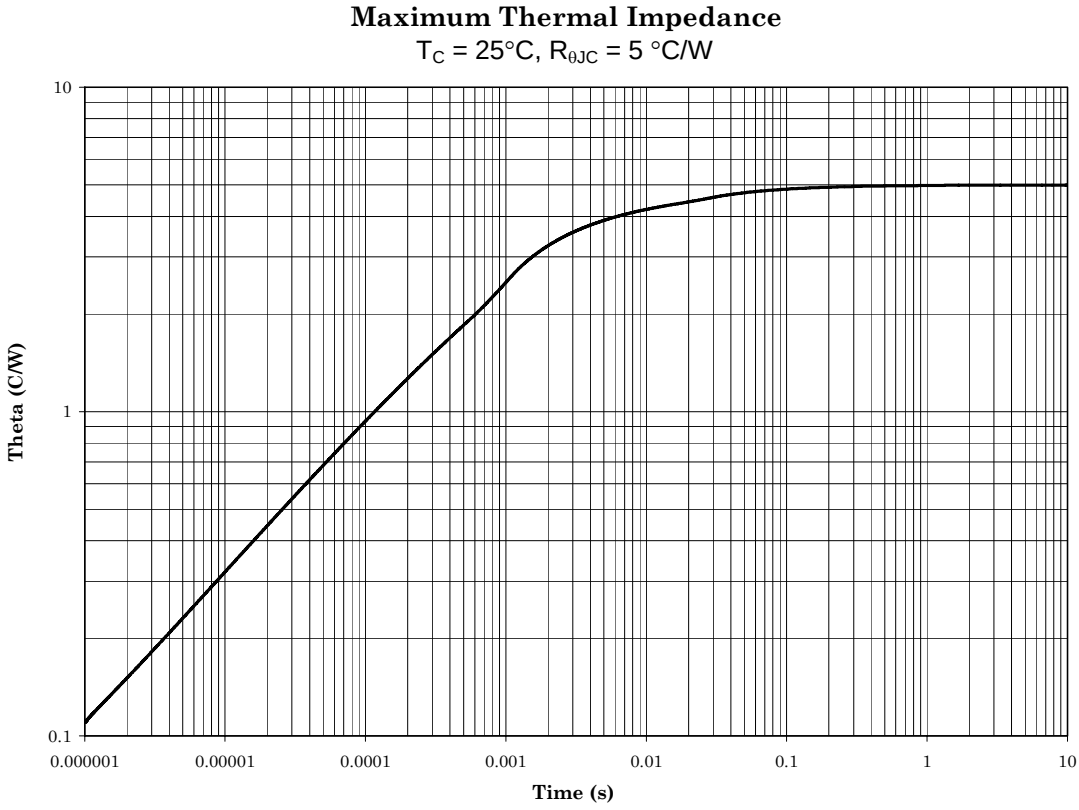


FIGURE 9. Thermal impedance for 2N5660 (similar to TO-66).

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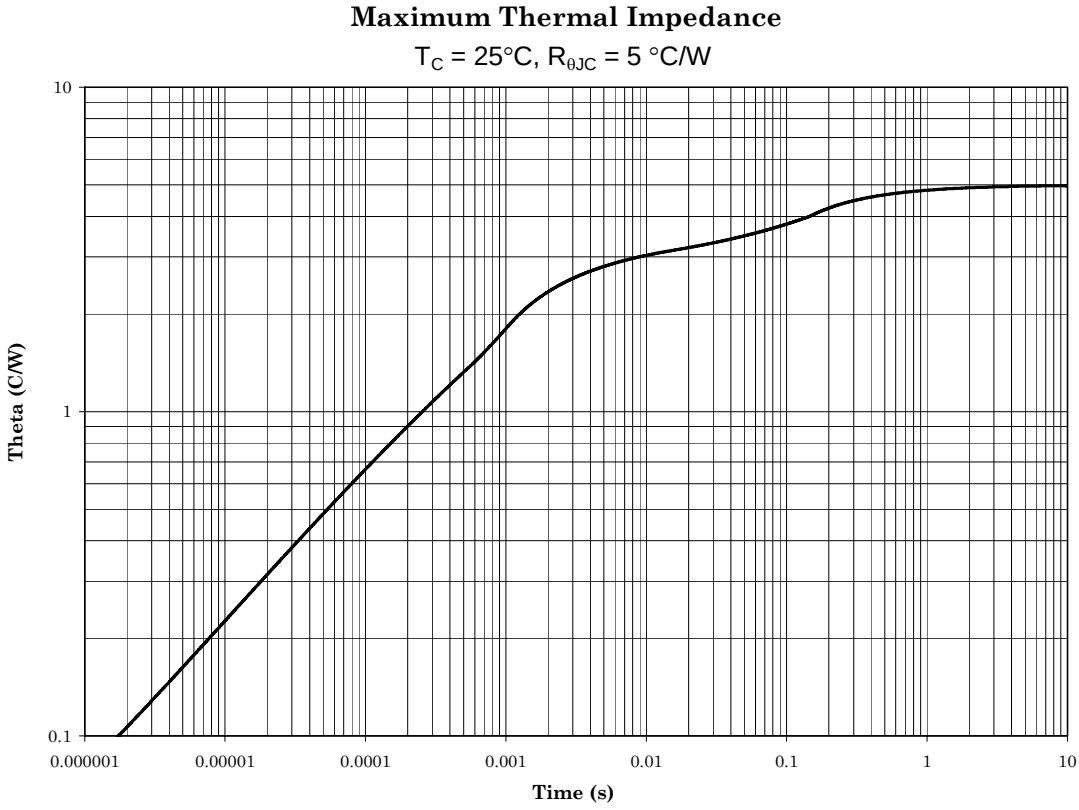


FIGURE 10. Thermal impedance for 2N5661 (similar to TO-66).

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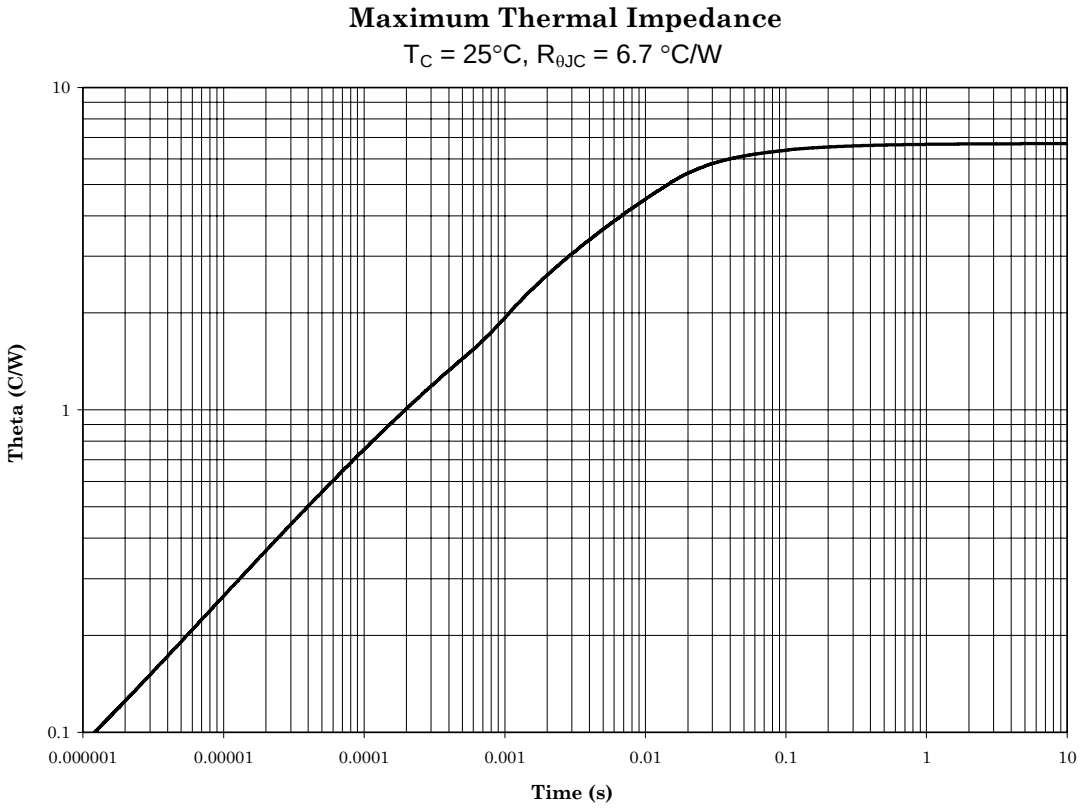


FIGURE 11. Thermal impedance for 2N5662 (similar to TO-5).

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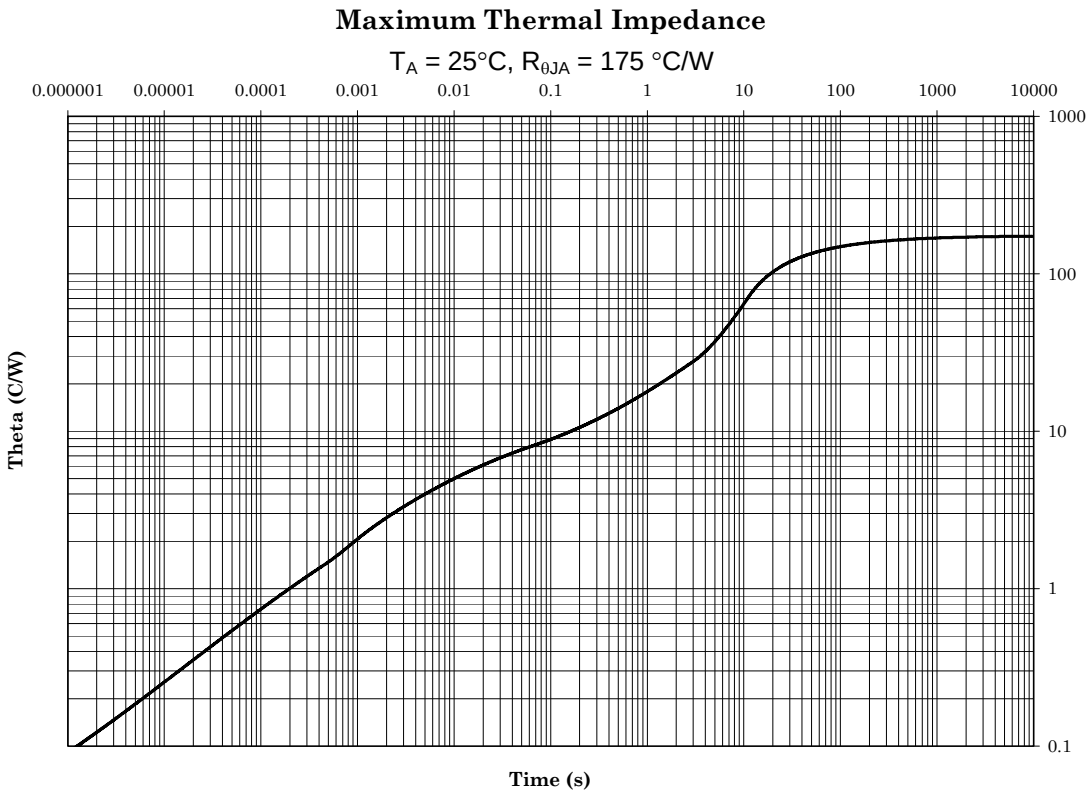


FIGURE 12. Thermal impedance for 2N5662 (similar to TO-5).

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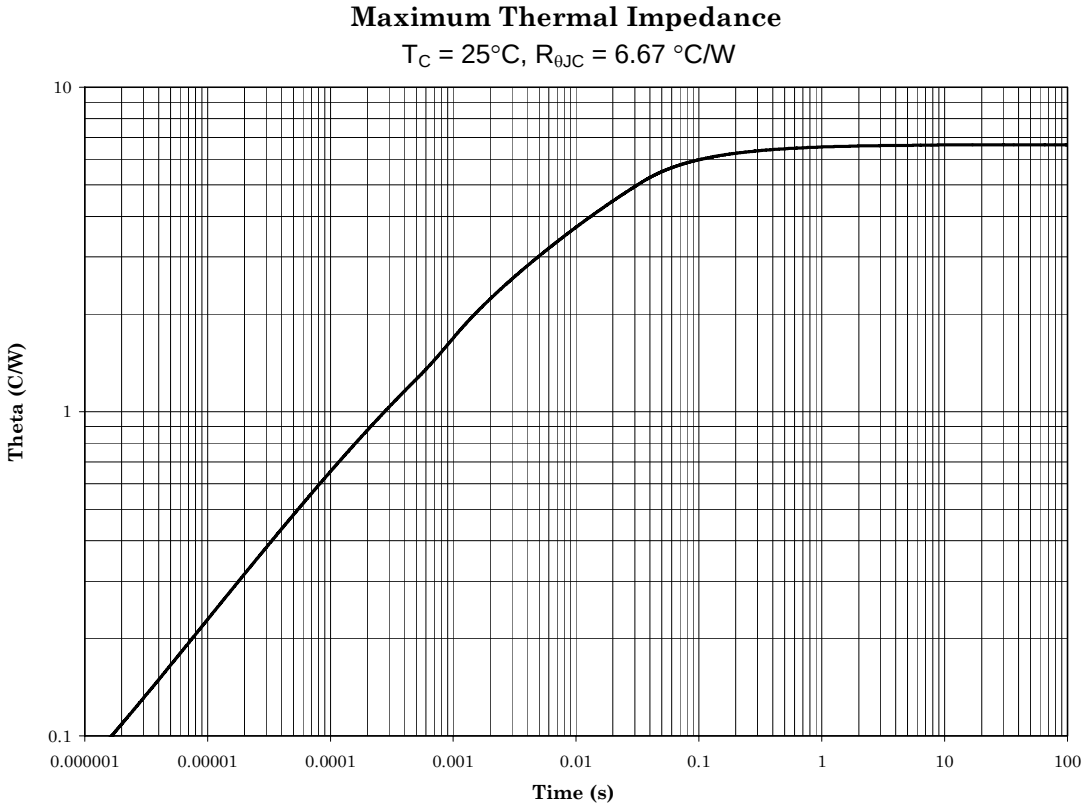


FIGURE 13. Thermal impedance for 2N5663 (similar to TO-5).

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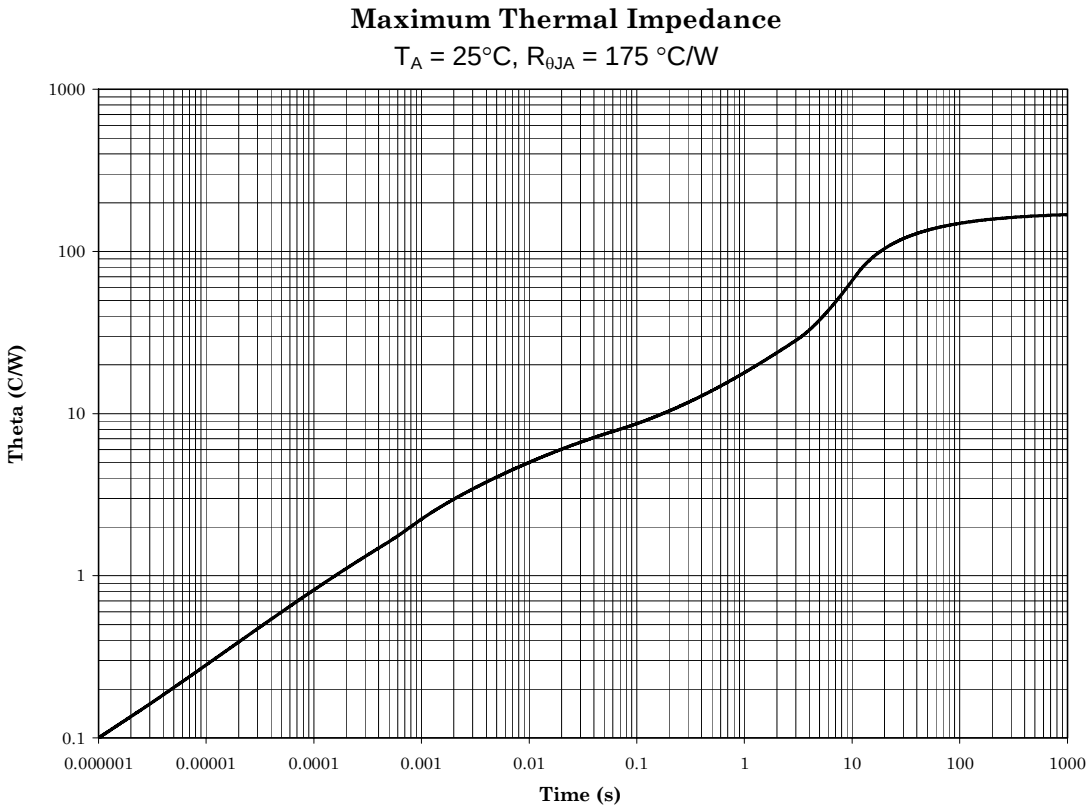


FIGURE 14. Thermal impedance for 2N5663 (similar to TO-5).

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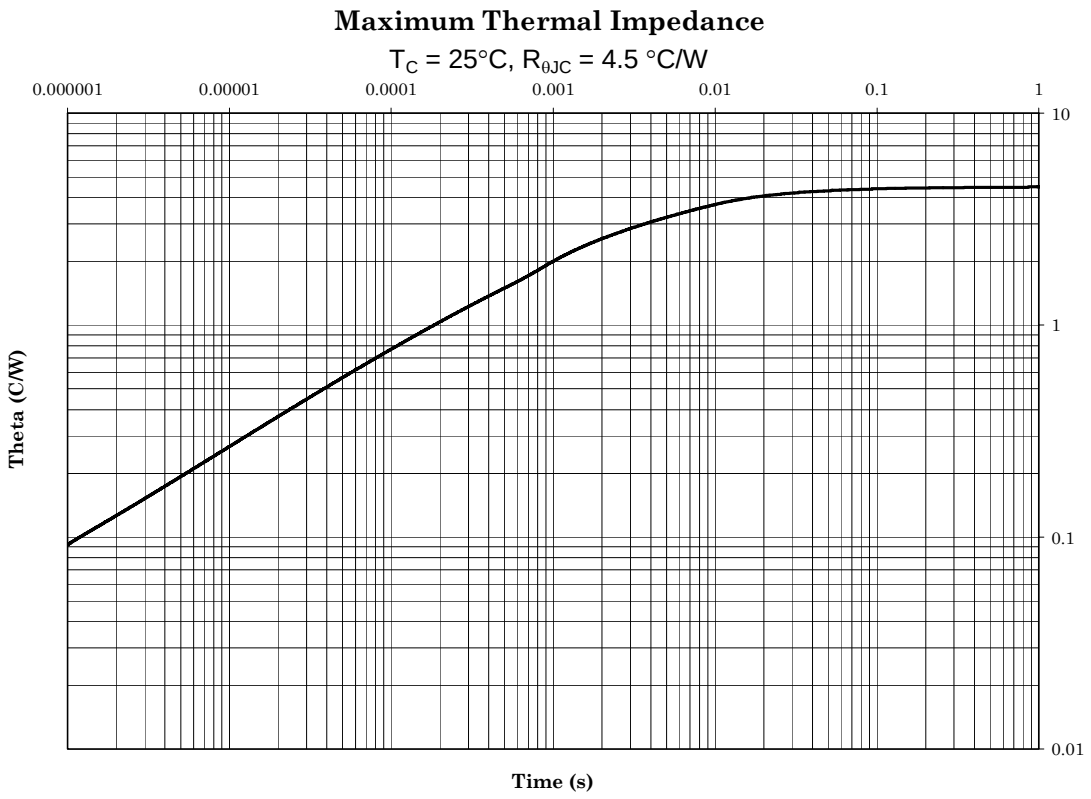


FIGURE 15. Thermal impedance for 2N5660U3 (U3).

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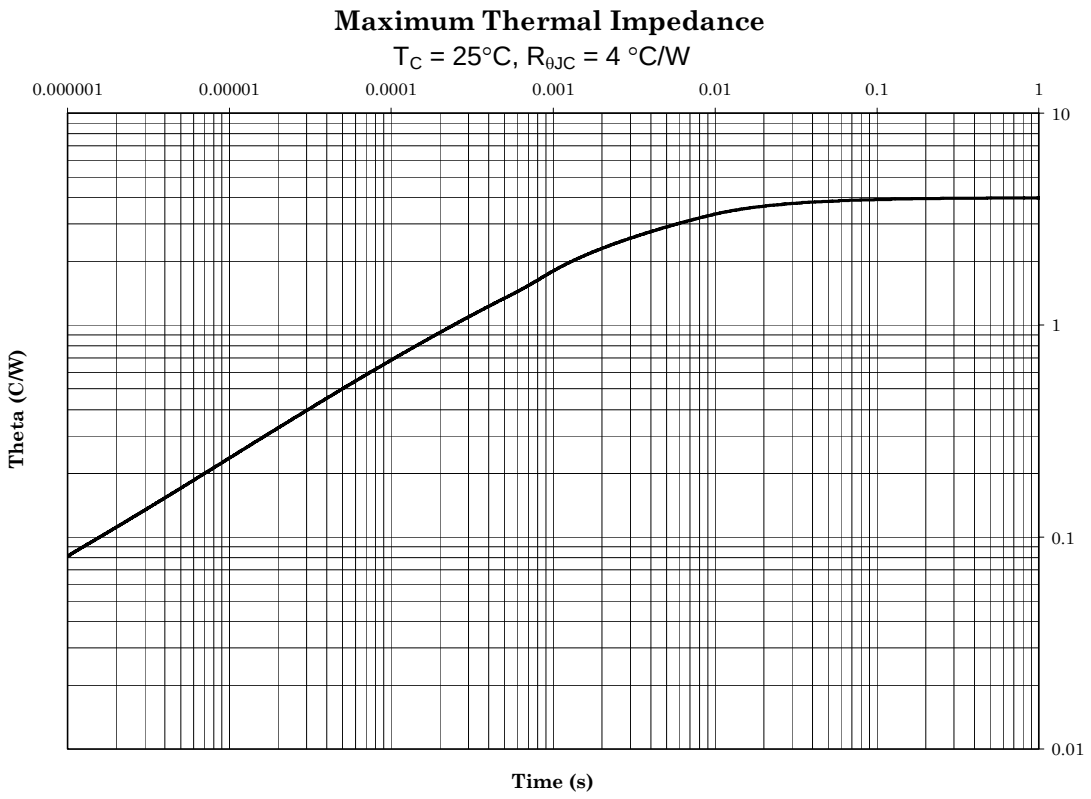
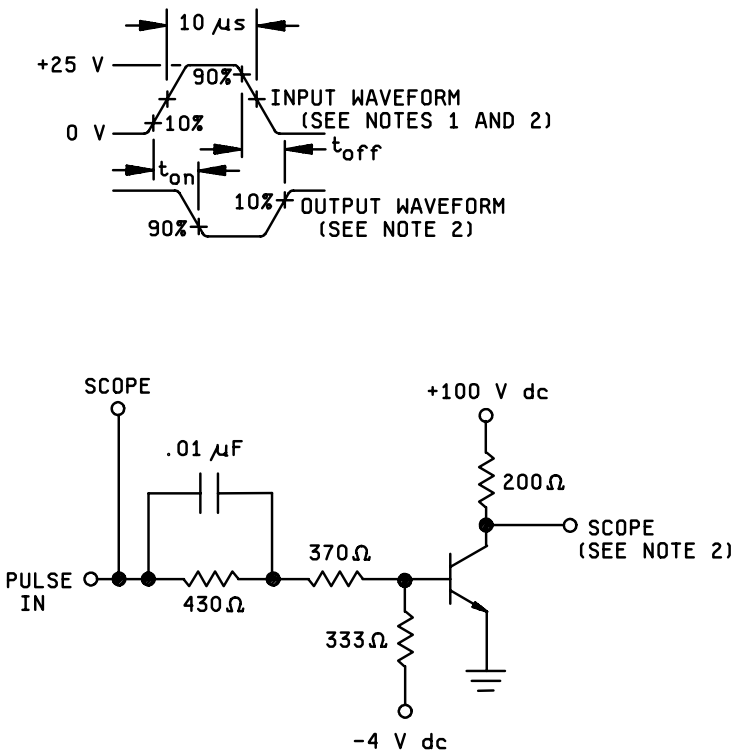


FIGURE 16. Thermal impedance for 2N5661U3 (U3).

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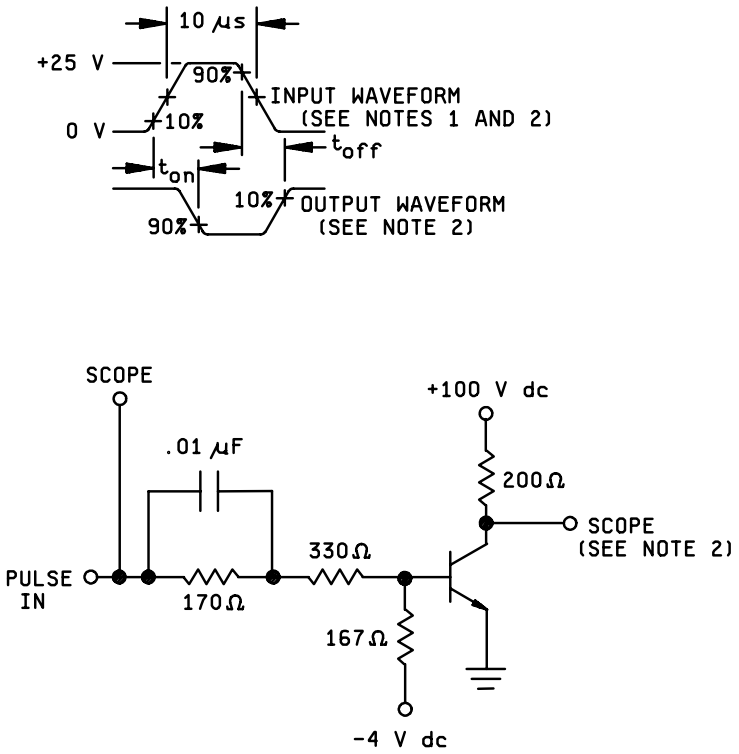


NOTES:

1. The input waveform is supplied by a pulse generator with the following characteristics: $t_r \leq 15$ ns, $t_f \leq 15$ ns, $Z_{OUT} = 50\ \Omega$, PW = 10 µs, duty cycle ≤ 2 percent.
2. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 1$ ns, $Z_{IN} \geq 10\ M\Omega$, $C_{IN} \leq 11.5$ pF.
3. Resistors shall be noninductive types.
4. The dc power supplies may require additional bypassing in order to minimize ringing.
5. The input pulse voltages, -4 V dc and +100 V dc, are nominal and shall be adjusted to obtain $I_{B1} = -I_{B2} = 15$ mA dc.
6. The 0.01 µF capacitor may be removed for current adjustment only.

FIGURE 17. Pulse response test circuit for types 2N5660 and 2N5662.

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NOTES:

1. The input waveform is supplied by a pulse generator with the following characteristics: $t_r \leq 15$ ns, $t_f \leq 15$ ns, $Z_{OUT} = 50\ \Omega$, PW = 10 µs, duty cycle ≤ 2 percent.
2. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 1$ ns, $Z_{IN} \geq 10\ M\Omega$, $C_{IN} \leq 11.5$ pF.
3. Resistors shall be noninductive types.
4. The dc power supplies may require additional by = passing in order to minimize ringing.
5. The input pulse voltages, -4 V dc and +100 V dc, are nominal and shall be adjusted to obtain $I_{B1} = -I_{B2} = 25$ mA dc.
6. The 0.01 µF capacitor may be removed for current adjustment only.

FIGURE 18. Pulse response test circuit for types 2N5661 and 2N5663.

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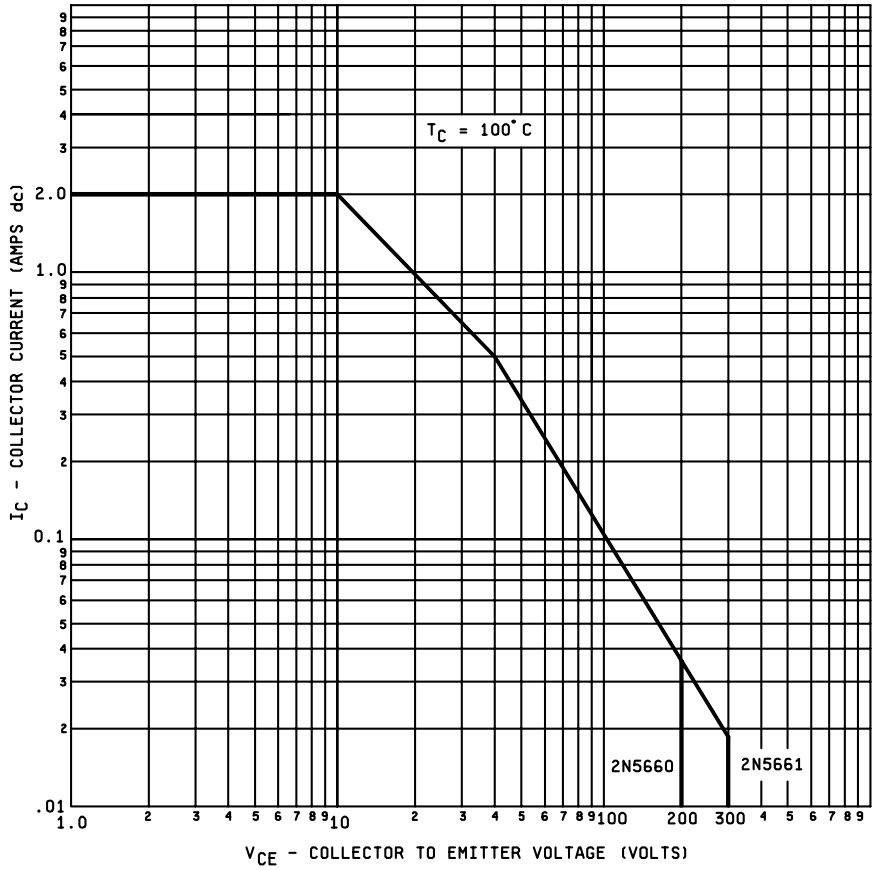


FIGURE 19. Maximum operating area graph (continuous dc) for types 2N5660 and 2N5661.

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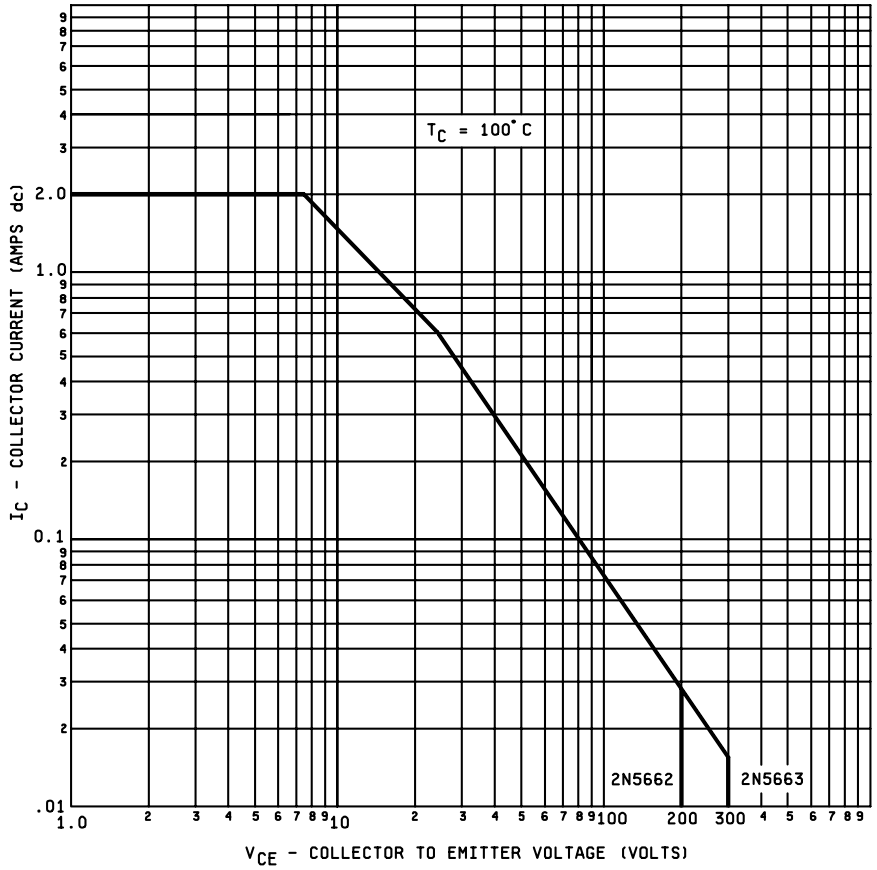


FIGURE 20. Maximum operating area graph (continuous dc) for types 2N5662 and 2N5663.

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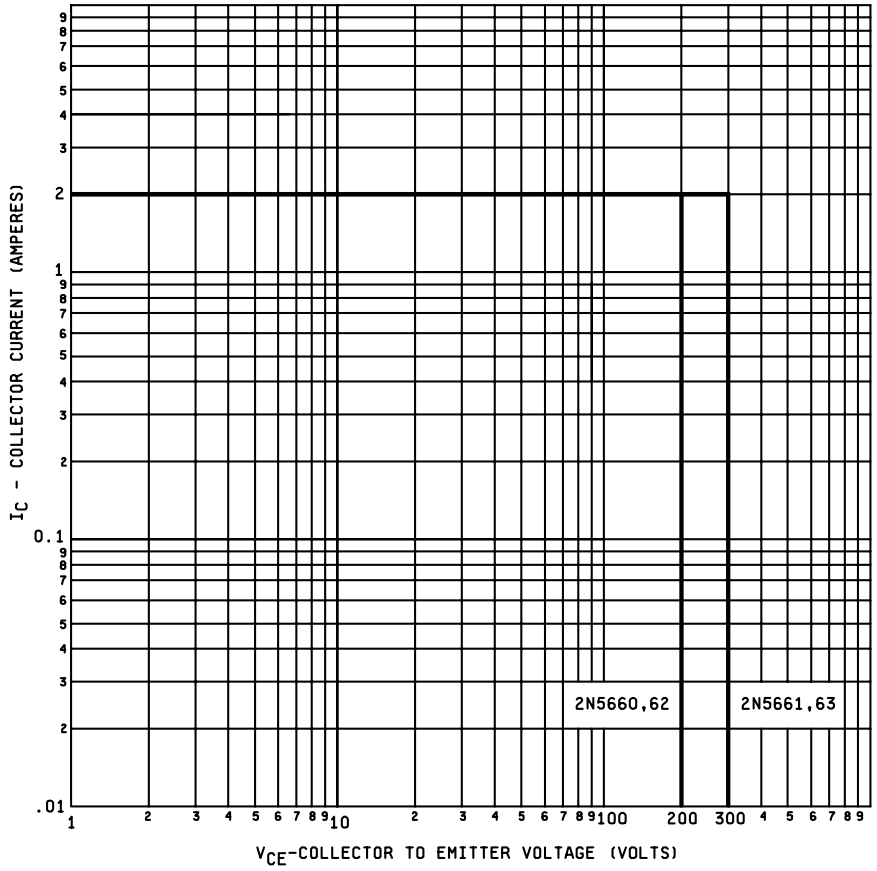
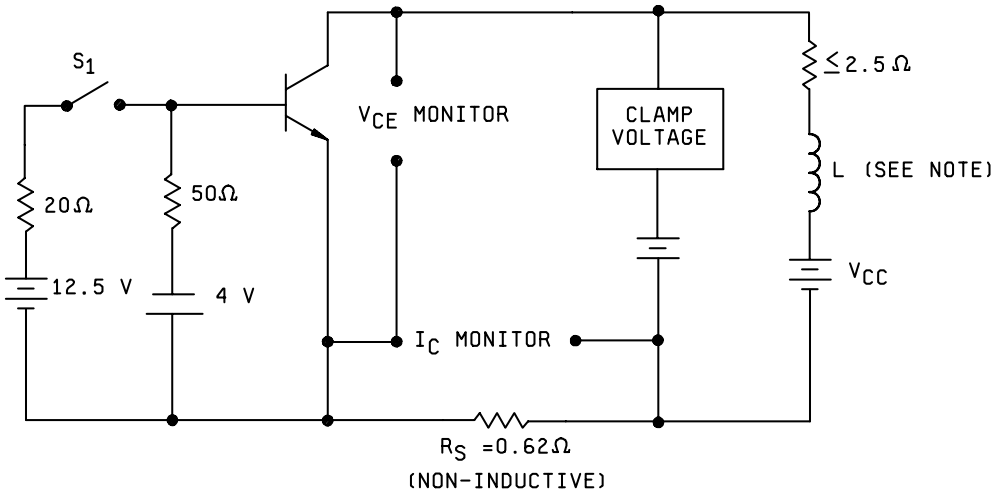


FIGURE 21. Safe operating area for switching between saturation and cutoff - (clamped inductive load).

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NOTE: 40 mH (triad C-48U, or equivalent)

Procedure:

1. With switch S_1 closed, set the specified test conditions.
2. Open S_1. Device fails if clamp voltage is not reached.
3. Perform specified end-point tests.

FIGURE 22. Clamped inductive sweep test circuit.

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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturer's List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC-VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

* 6.4 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's Part or Identifying Number (PIN). This information in no way implies that manufacturer's PIN's are suitable for the military PIN. The term PIN is equivalent to the term (part number, identification number, and type designator) which was previously used in this specification.

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* 6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2711)

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil/.

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